

Chapter 5

Association Analysis: Basic Concepts

Introduction to Data Mining, 2nd Edition

by

Tan, Steinbach, Karpatne, Kumar

Association Rule Mining

- Given a set of transactions, find rules that will predict the occurrence of an item based on the occurrences of other items in the transaction

Market-Basket transactions

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Association Rules

$\{\text{Diaper}\} \rightarrow \{\text{Beer}\},$
 $\{\text{Milk, Bread}\} \rightarrow \{\text{Eggs, Coke}\},$
 $\{\text{Beer, Bread}\} \rightarrow \{\text{Milk}\},$

Implication means **co-occurrence**,
not causality!

Definition: Frequent Itemset

- **Itemset**

- A collection of one or more items
 - ◆ Example: {Milk, Bread, Diaper}
- k-itemset
 - ◆ An itemset that contains k items

- **Support count (σ)**

- Frequency of occurrence of an itemset
- E.g. $\sigma(\{\text{Milk, Bread, Diaper}\}) = 2$

- **Support**

- Fraction of transactions that contain an itemset
- E.g. $s(\{\text{Milk, Bread, Diaper}\}) = 2/5$

- **Frequent Itemset**

- An itemset whose support is greater than or equal to a *minsup* threshold

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Definition: Association Rule

- **Association Rule**

- An implication expression of the form $X \rightarrow Y$, where X and Y are itemsets
- Example:
 $\{\text{Milk, Diaper}\} \rightarrow \{\text{Beer}\}$

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
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4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

- **Rule Evaluation Metrics**

- Support (s)
 - ◆ Fraction of transactions that contain both X and Y
- Confidence (c)
 - ◆ Measures how often items in Y appear in transactions that contain X

Example:

$\{\text{Milk, Diaper}\} \Rightarrow \{\text{Beer}\}$

$$s = \frac{\sigma(\text{Milk, Diaper, Beer})}{|T|} = \frac{2}{5} = 0.4$$

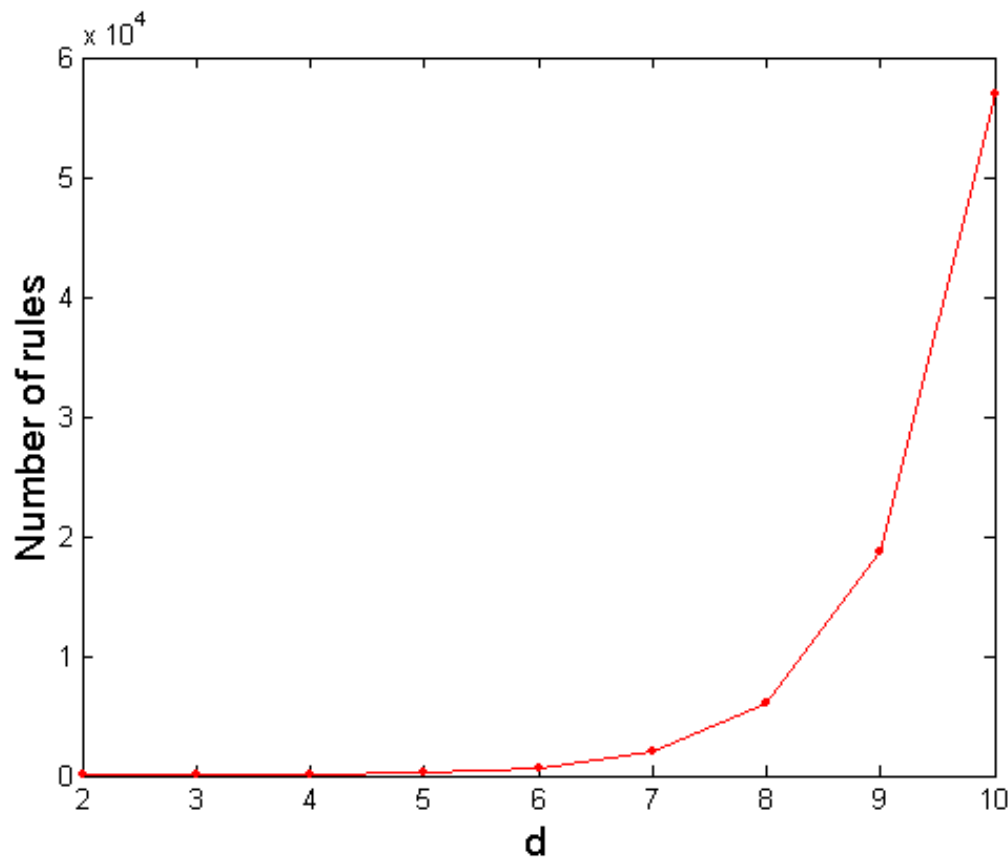
$$c = \frac{\sigma(\text{Milk, Diaper, Beer})}{\sigma(\text{Milk, Diaper})} = \frac{2}{3} = 0.67$$

Association Rule Mining Task

- Given a set of transactions T , the goal of association rule mining is to find all rules having
 - support $\geq \textit{minsup}$ threshold
 - confidence $\geq \textit{minconf}$ threshold
 - Brute-force approach:
 - List all possible association rules
 - Compute the support and confidence for each rule
 - Prune rules that fail the *minsup* and *minconf* thresholds
- ⇒ **Computationally prohibitive!**

Computational Complexity

- Given d unique items:
 - Total number of itemsets = 2^d
 - Total number of possible association rules:



$$R = \sum_{k=1}^{d-1} \left[\binom{d}{k} \times \sum_{j=1}^{d-k} \binom{d-k}{j} \right]$$
$$= 3^d - 2^{d+1} + 1$$

If $d=6$, $R = 602$ rules

Mining Association Rules

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Rules:

$\{\text{Milk, Diaper}\} \rightarrow \{\text{Beer}\}$ ($s=0.4$, $c=0.67$)
 $\{\text{Milk, Beer}\} \rightarrow \{\text{Diaper}\}$ ($s=0.4$, $c=1.0$)
 $\{\text{Diaper, Beer}\} \rightarrow \{\text{Milk}\}$ ($s=0.4$, $c=0.67$)
 $\{\text{Beer}\} \rightarrow \{\text{Milk, Diaper}\}$ ($s=0.4$, $c=0.67$)
 $\{\text{Diaper}\} \rightarrow \{\text{Milk, Beer}\}$ ($s=0.4$, $c=0.5$)
 $\{\text{Milk}\} \rightarrow \{\text{Diaper, Beer}\}$ ($s=0.4$, $c=0.5$)

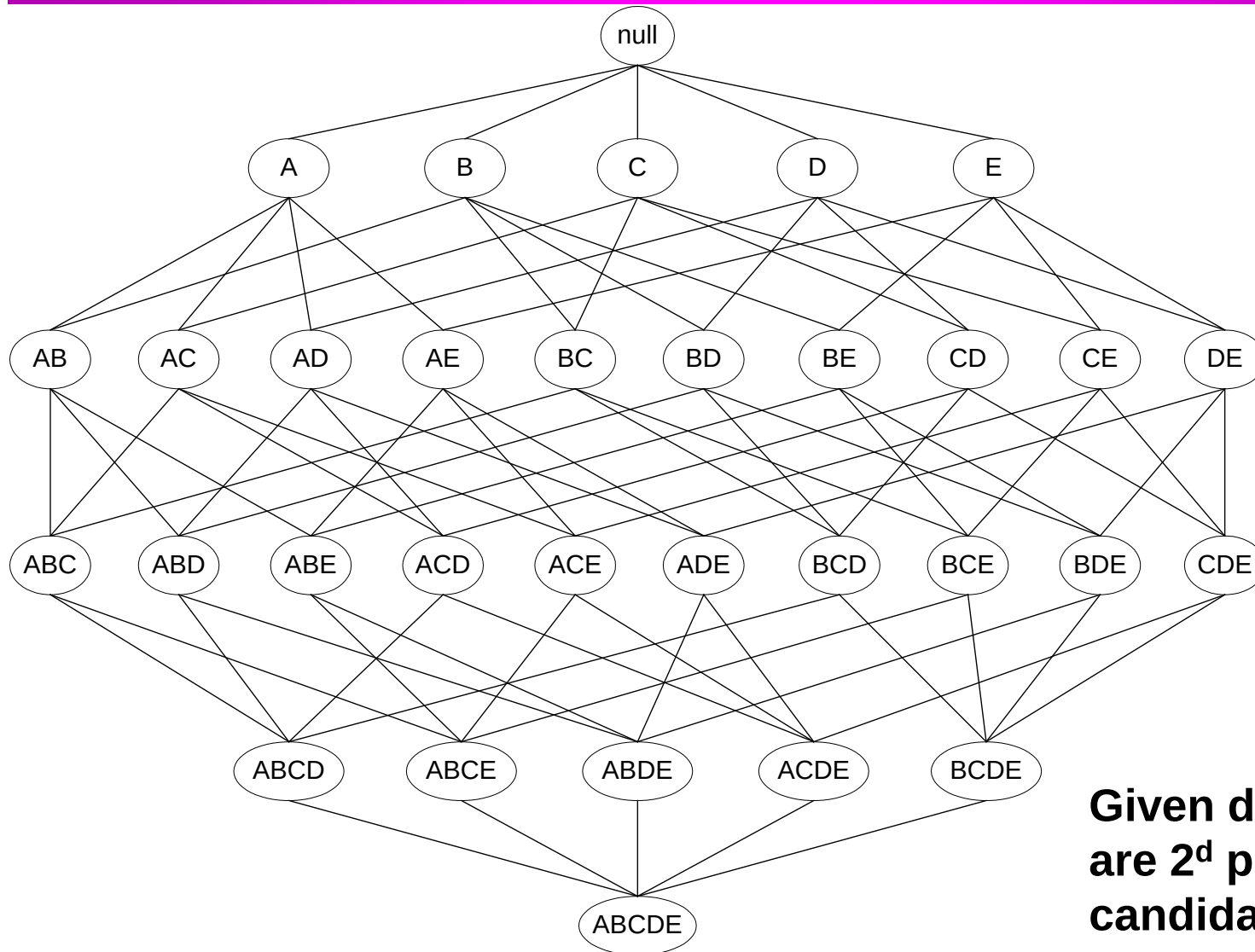
Observations:

- All the above rules are binary partitions of the same itemset:
 $\{\text{Milk, Diaper, Beer}\}$
- Rules originating from the same itemset have identical support but can have different confidence
- Thus, we may decouple the support and confidence requirements

Mining Association Rules

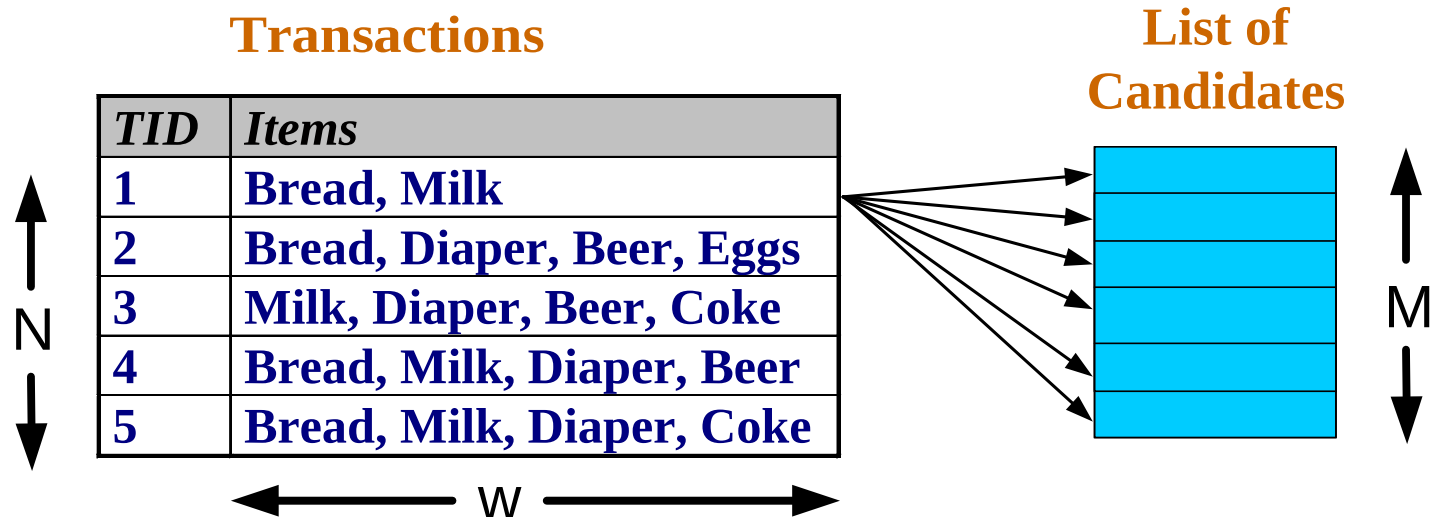
- Two-step approach:
 1. Frequent Itemset Generation
 - Generate all itemsets whose support $\geq$ minsup
 2. Rule Generation
 - Generate high confidence rules from each frequent itemset, where each rule is a binary partitioning of a frequent itemset
- Frequent itemset generation is still computationally expensive

Frequent Itemset Generation



Frequent Itemset Generation

- Brute-force approach:
 - Each itemset in the lattice is a **candidate** frequent itemset
 - Count the support of each candidate by scanning the database



- Match each transaction against every candidate
- Complexity $\sim O(NMw) \Rightarrow$ **Expensive since $M = 2^d$!!!**

Frequent Itemset Generation Strategies

- Reduce the **number of candidates** (M)
 - Complete search: $M=2^d$
 - Use pruning techniques to reduce M
- Reduce the **number of transactions** (N)
 - Reduce size of N as the size of itemset increases
 - Used by DHP and vertical-based mining algorithms
- Reduce the **number of comparisons** (NM)
 - Use efficient data structures to store the candidates or transactions
 - No need to match every candidate against every transaction

Reducing Number of Candidates

- **Apriori principle:**

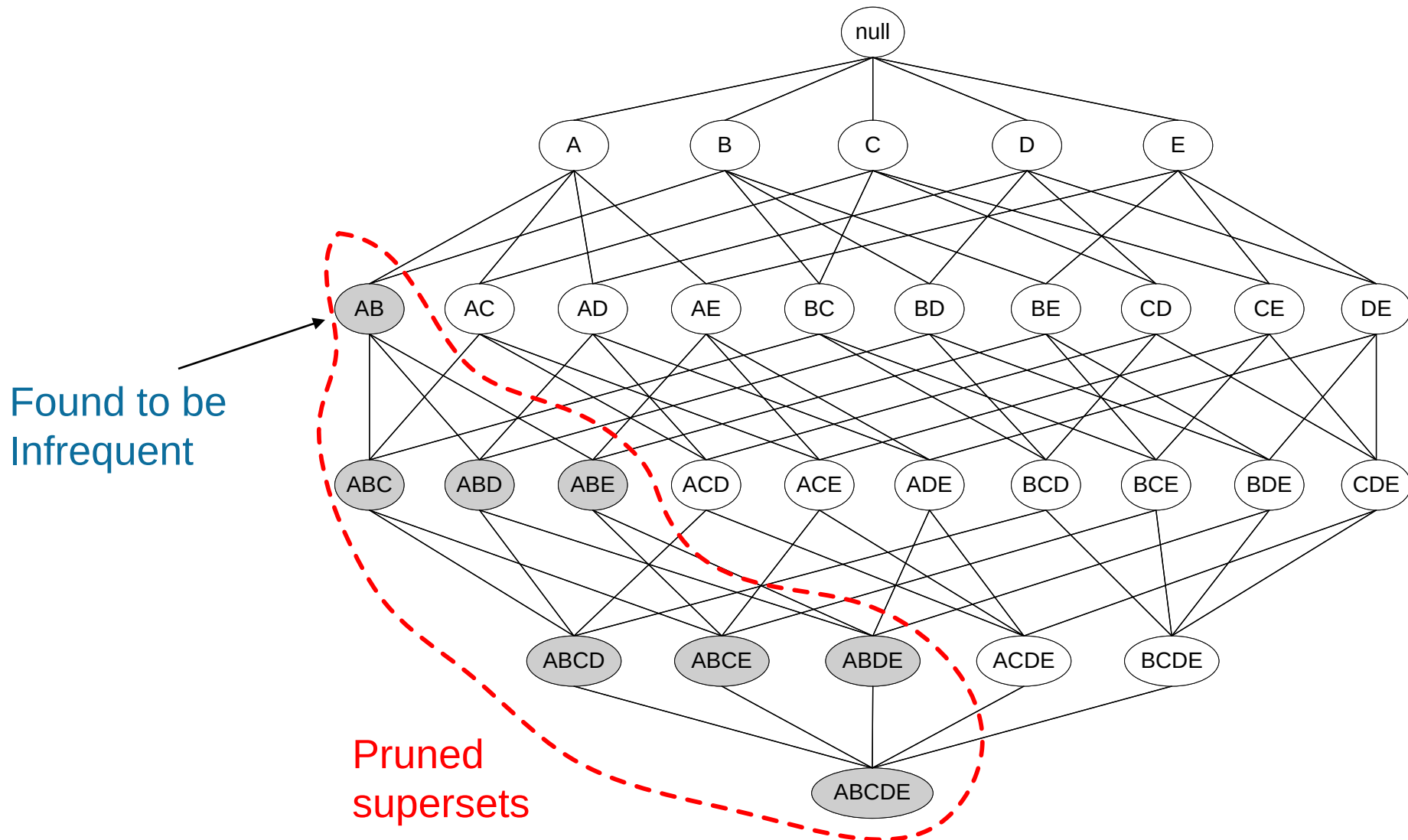
- If an itemset is frequent, then **all of its subsets** must also be frequent

- Apriori principle holds due to the following property of the support measure:

$$\forall X, Y : (X \subseteq Y) \Rightarrow s(X) \geq s(Y)$$

- Support of an itemset never exceeds the support of its subsets
- This is known as the **anti-monotone** property of support

Illustrating Apriori Principle



Illustrating Apriori Principle

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Beer, Bread, Diaper, Eggs
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Bread, Coke, Diaper, Milk



Items (1-itemsets)

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Minimum Support = 3

If every subset is considered,

$${}^6C_1 + {}^6C_2 + {}^6C_3$$

$$6 + 15 + 20 = 41$$

With support-based pruning,

$$6 + 6 + 4 = 16$$

Illustrating Apriori Principle

TID	Items
1	Bread, Milk
2	Beer, Bread, Diaper, Eggs
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Bread, Coke, Diaper, Milk

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Items (1-itemsets)



Itemset	Count
{Bread, Milk}	3
{Beer, Bread}	2
{Bread, Diaper}	3
{Beer, Milk}	2
{Diaper, Milk}	3
{Beer, Diaper}	3

Pairs (2-itemsets)

(No need to generate candidates involving Coke or Eggs)

Minimum Support = 3

If every subset is considered,

$${}^6C_1 + {}^6C_2 + {}^6C_3$$

$$6 + \mathbf{15} + 20 = 41$$

With support-based pruning,

$${}^6C_1 + {}^4C_2 + {}^4C_3$$

$$6 + \mathbf{6} + 4 = 16$$

Illustrating Apriori Principle

TID	Items
1	Bread, Milk
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Item	Count
Bread	4
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Milk	4
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Diaper	4
Eggs	1

Items (1-itemsets)



Itemset	Count
{Bread, Milk}	3
{Bread, Beer}	2
{Bread, Diaper}	3
{Milk, Beer}	2
{Milk, Diaper}	3
{Beer, Diaper}	3

Pairs (2-itemsets)



Triplets (3-itemsets)

Itemset	Count
{ Beer, Diaper, Milk}	2
{ Beer, Bread, Diaper}	2
{Bread, Diaper, Milk}	2
{Beer, Bread, Milk}	1

Minimum Support = 3

If every subset is considered,

$${}^6C_1 + {}^6C_2 + {}^6C_3$$

$$6 + 15 + \mathbf{20} = 41$$

With support-based pruning,

$${}^6C_1 + {}^4C_2 + {}^4C_3$$

$$6 + 6 + \mathbf{4} = 16$$

Apriori Algorithm

- F_k : frequent k-itemsets
- L_k : candidate k-itemsets
- Algorithm
 - Let $k=1$
 - Generate $F_1 = \{\text{frequent 1-itemsets}\}$
 - Repeat until F_k is empty
 - ◆ **Candidate Generation:** Generate L_{k+1} from F_k
 - ◆ **Candidate Pruning:** Prune candidate itemsets in L_{k+1} containing subsets of length k that are infrequent
 - ◆ **Support Counting:** Count the support of each candidate in L_{k+1} by scanning the DB
 - ◆ **Candidate Elimination:** Eliminate candidates in L_{k+1} that are infrequent, leaving only those that are frequent $\Rightarrow F_{k+1}$

Candidate Generation: $F_{k-1} \times F_{k-1}$ Method

- Merge two frequent $(k-1)$ -itemsets if their first $(k-2)$ items are identical
- $F_3 = \{ABC, ABD, ABE, ACD, BCD, BDE, CDE\}$
 - Merge(ABC, ABD) = ABCD
 - Merge(ABC, ABE) = ABCE
 - Merge(ABD, ABE) = ABDE
 - Do not merge(ABD, ACD) because they share only prefix of length 1 instead of length 2

Illustrating Apriori Principle

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Items (1-itemsets)



Itemset	Count
{Bread,Milk}	3
{Bread,Beer}	2
{Bread,Diaper}	3
{Milk,Beer}	2
{Milk,Diaper}	3
{Beer,Diaper}	3

Pairs (2-itemsets)

(No need to generate candidates involving Coke or Eggs)

Minimum Support = 3

If every subset is considered,

$${}^6C_1 + {}^6C_2 + {}^6C_3 \\ 6 + 15 + 20 = 41$$

With support-based pruning,

$$6 + 6 + 1 = 13$$



Itemset	Count
{Bread, Diaper, Milk}	2

Triplets (3-itemsets)

Use of $F_{k-1} \times F_{k-1}$ method for candidate generation results in only one 3-itemset. This is eliminated after the support counting step.

Candidate Pruning

- Let $F_3 = \{ABC, ABD, ABE, ACD, BCD, BDE, CDE\}$ be the set of frequent 3-itemsets
- $L_4 = \{ABCD, ABCE, ABDE\}$ is the set of candidate 4-itemsets generated (from previous slide)
- Candidate pruning
 - Prune ABCE because ACE and BCE are infrequent
 - Prune ABDE because ADE is infrequent
- After candidate pruning: $L_4 = \{ABCD\}$

Illustrating Apriori Principle

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Item	Count
Bread	4
Coke	2
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Beer	3
Diaper	4
Eggs	1

Items (1-itemsets)



Itemset	Count
{Bread, Milk}	3
{Bread, Beer}	2
{Bread, Diaper}	3
{Milk, Beer}	2
{Milk, Diaper}	3
{Beer, Diaper}	3

Pairs (2-itemsets)

(No need to generate candidates involving Coke or Eggs)

Minimum Support = 3

If every subset is considered,

$${}^6C_1 + {}^6C_2 + {}^6C_3 \\ 6 + 15 + 20 = 41$$

With support-based pruning,

$$6 + 6 + 4 = 16 \\ 6 + 6 + 1 = 13$$



Triplets (3-itemsets)

Itemset	Count
{ Beer, Diaper, Milk}	2
{ Beer, Bread, Diaper}	2
{Bread, Diaper, Milk}	2
{Beer, Bread, Milk}	1

Support Counting of Candidate Itemsets

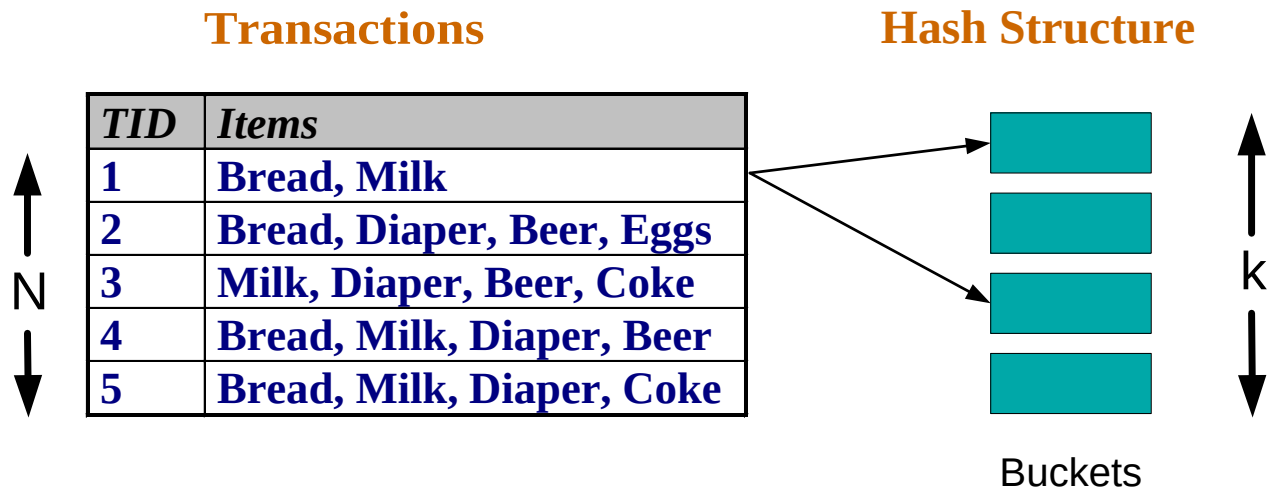
- Scan the database of transactions to determine the support of each candidate itemset
 - Must match every candidate itemset against every transaction, which is an expensive operation

<i>TID</i>	<i>Items</i>
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5	Bread, Coke, Diaper, Milk

Itemset
{ Beer, Diaper, Milk}
{ Beer,Bread,Diaper}
{Bread, Diaper, Milk}
{ Beer, Bread, Milk}

Support Counting of Candidate Itemsets

- To reduce number of comparisons, store the candidate itemsets in a hash structure
 - Instead of matching each transaction against every candidate, match it against candidates contained in the hashed buckets

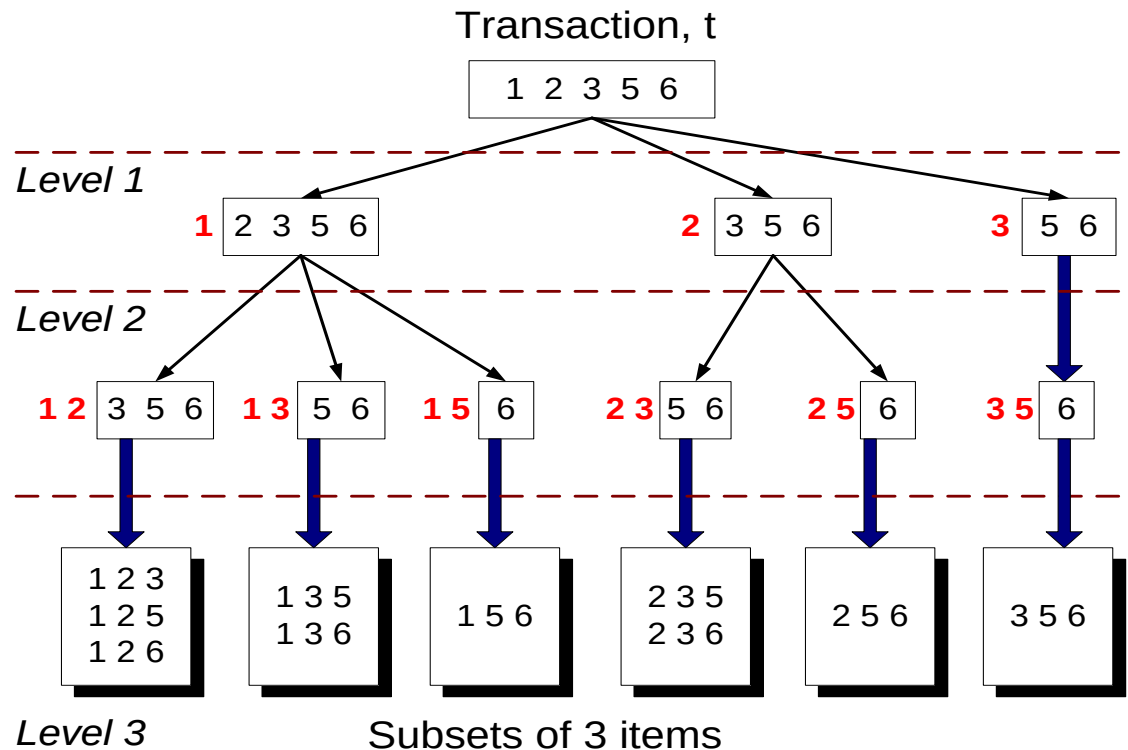


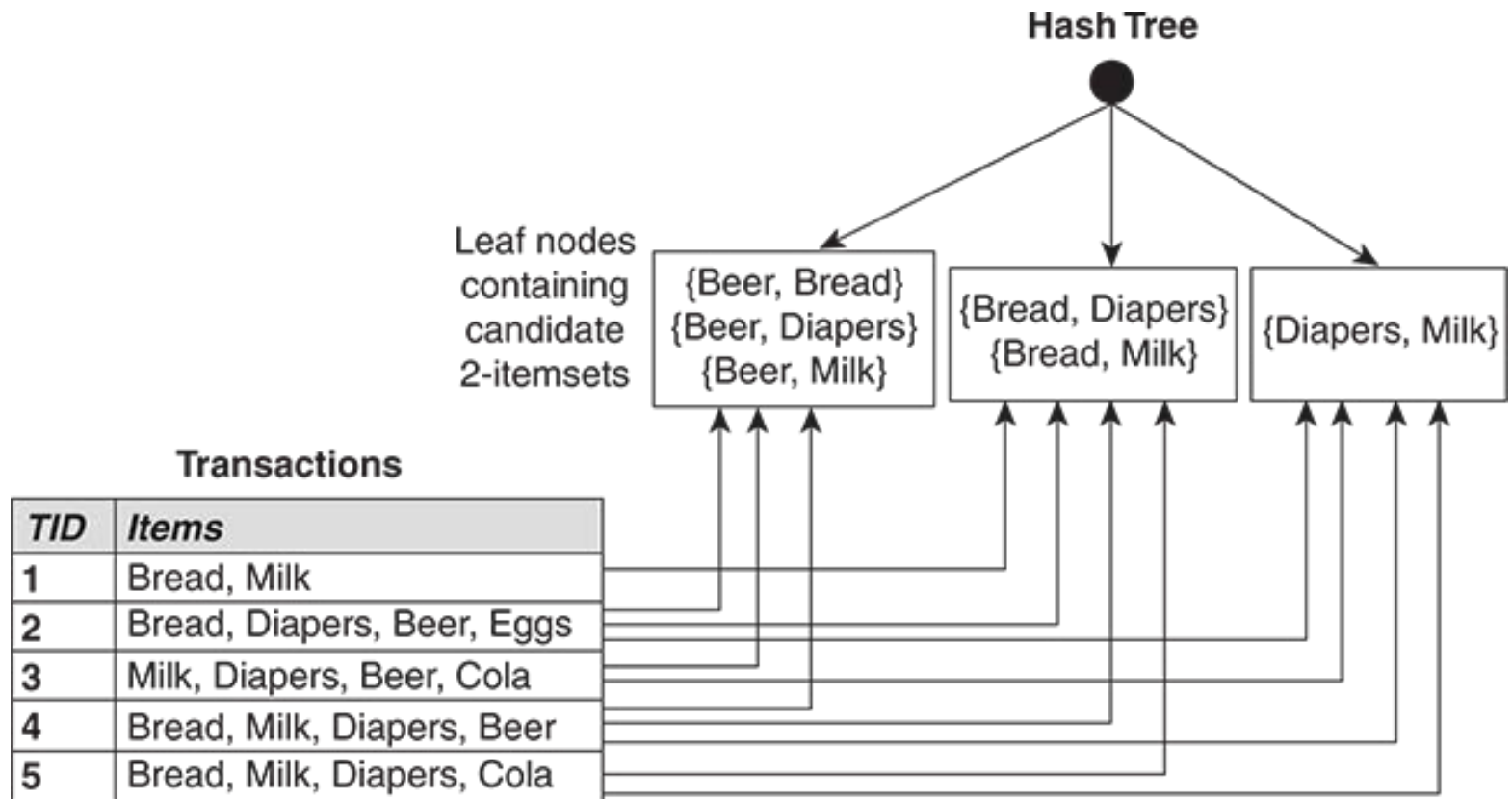
Support Counting: An Example

Suppose you have 15 candidate itemsets of length 3:

{1 4 5}, {1 2 4}, {4 5 7}, {1 2 5}, {4 5 8}, {1 5 9}, {1 3 6}, {2 3 4}, {5 6 7}, {3 4 5},
{3 5 6}, {3 5 7}, {6 8 9}, {3 6 7}, {3 6 8}

How many of these itemsets are supported by transaction (1,2,3,5,6)?





Rule Generation

- Given a frequent itemset L , find all non-empty subsets $f \subset L$ such that $f \rightarrow L - f$ satisfies the minimum confidence requirement
 - If $\{A,B,C,D\}$ is a frequent itemset, candidate rules:

$ABC \rightarrow D,$	$ABD \rightarrow C,$	$ACD \rightarrow B,$	$BCD \rightarrow A,$
$A \rightarrow BCD,$	$B \rightarrow ACD,$	$C \rightarrow ABD,$	$D \rightarrow ABC$
$AB \rightarrow CD,$	$AC \rightarrow BD,$	$AD \rightarrow BC,$	$BC \rightarrow AD,$
$BD \rightarrow AC,$	$CD \rightarrow AB,$		
- If $|L| = k$, then there are $2^k - 2$ candidate association rules (ignoring $L \rightarrow \emptyset$ and $\emptyset \rightarrow L$)

Rule Generation

- In general, confidence **does not** have an anti-monotone property

$c(ABC \rightarrow D)$ can be larger or smaller than $c(AB \rightarrow D)$

- But confidence of rules generated from the same itemset has an anti-monotone property

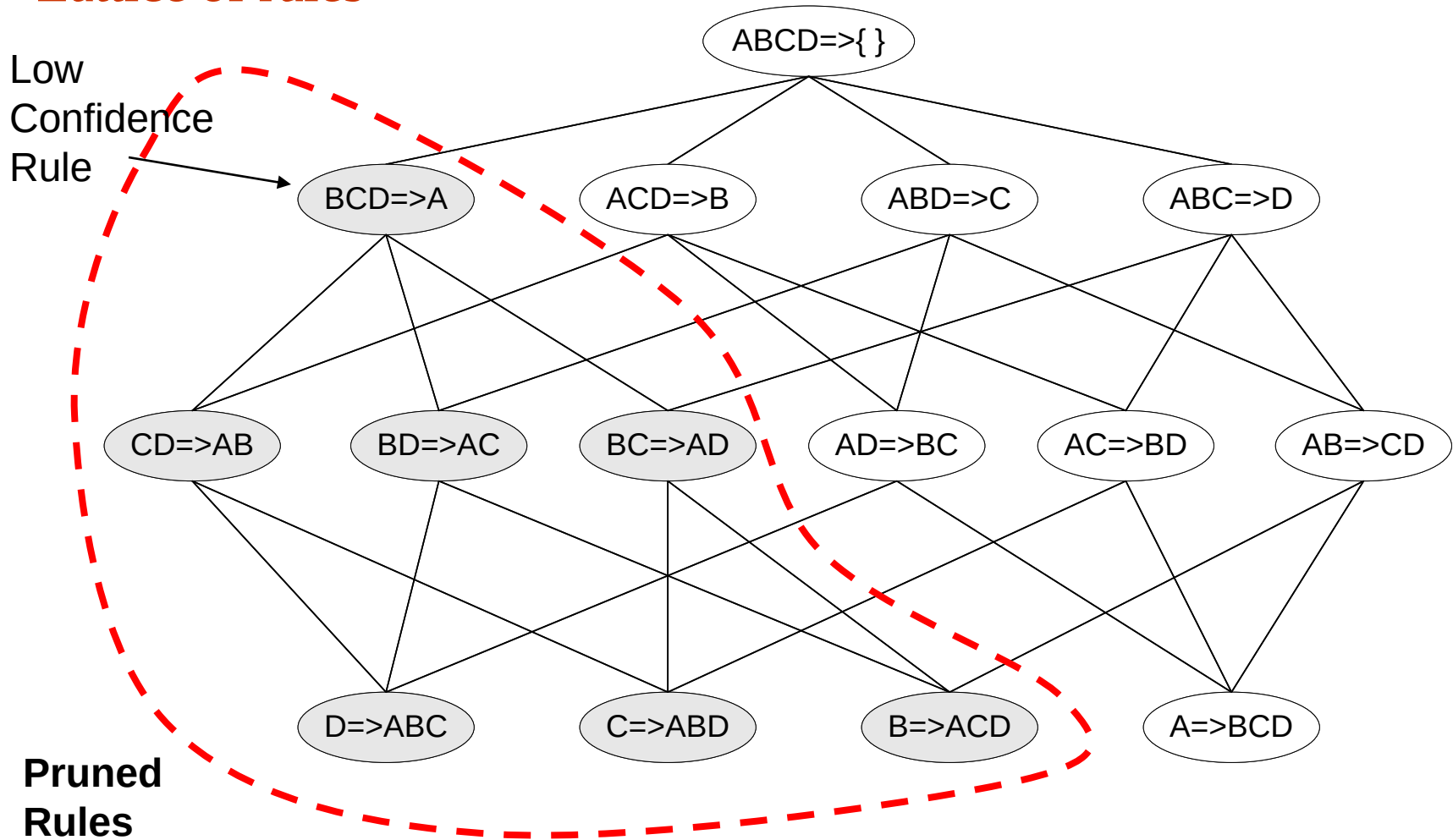
- E.g., Suppose $\{A, B, C, D\}$ is a frequent 4-itemset:

$$c(ABC \rightarrow D) \geq c(AB \rightarrow CD) \geq c(A \rightarrow BCD)$$

- Confidence is anti-monotone w.r.t. number of items on the RHS of the rule

Rule Generation for Apriori Algorithm

Lattice of rules



Association Analysis: Basic Concepts and Algorithms

Algorithms and Complexity

Factors Affecting Complexity of Apriori

- Choice of minimum support threshold
 - lowering support threshold results in more frequent itemsets
 - this may increase number of candidates and max length of frequent itemsets
- Dimensionality (number of items) of the data set
 - More space is needed to store support count of itemsets
 - if number of frequent itemsets also increases, both computation and I/O costs may also increase
- Size of database
 - run time of algorithm increases with number of transactions
- Average transaction width
 - transaction width increases the max length of frequent itemsets
 - number of subsets in a transaction increases with its width, increasing computation time for support counting

Pattern Evaluation

- Association rule algorithms can produce large number of rules
- Interestingness measures can be used to prune/rank the patterns
 - In the original formulation, support & confidence are the only measures used

Computing Interestingness Measure

- Given $X \rightarrow Y$ or $\{X, Y\}$, information needed to compute interestingness can be obtained from a contingency table

Contingency table

	Y	$\overline{Y}$	
X	f_{11}	f_{10}	f_{1+}
$\overline{X}$	f_{01}	f_{00}	f_{0+}
	f_{+1}	f_{+0}	N

f_{11} : support of X and Y

f_{10} : support of $\underline{X}$ and $\overline{Y}$

f_{01} : support of $\overline{X}$ and $\underline{Y}$

f_{00} : support of $\overline{X}$ and $\overline{Y}$

Used to define various measures

- ◆ support, confidence, Gini, entropy, etc.

Drawback of Confidence

Customers	Tea	Coffee	...
C1	0	1	...
C2	1	0	...
C3	1	1	...
C4	1	0	...
...			

	<i>Coffee</i>	$\overline{Coffee}$	
<i>Tea</i>	150	50	200
$\overline{Tea}$	650	150	800
	800	200	1000

Association Rule: Tea $\rightarrow$ Coffee

Confidence $\cong P(\text{Coffee}|\text{Tea}) = 150/200 = 0.75$

Confidence > 50%, meaning people who drink tea are more likely to drink coffee than not drink coffee

So rule seems reasonable

Drawback of Confidence

	Coffee	<u>Coffee</u>	
Tea	150	50	200
<u>Tea</u>	650	150	800
	800	200	1000

Association Rule: Tea $\rightarrow$ Coffee

Confidence = $P(\text{Coffee}|\text{Tea}) = 150/200 = 0.75$

but $P(\text{Coffee}) = 0.8$, which means knowing that a person drinks tea reduces the probability that the person drinks coffee!

$\Rightarrow$ Note that $P(\text{Coffee}|\overline{\text{Tea}}) = 650/800 = 0.8125$

Drawback of Confidence

Custo mers	Tea	Honey	...
C1	0	1	...
C2	1	0	...
C3	1	1	...
C4	1	0	...
...			

	<i>Honey</i>	$\overline{Honey}$	
<i>Tea</i>	100	100	200
$\overline{Tea}$	20	780	800
	120	880	1000

Association Rule: Tea $\rightarrow$ Honey

Confidence $\cong P(\text{Honey}|\text{Tea}) = 100/200 = 0.50$

Confidence = 50%, which may mean that drinking tea has little influence whether honey is used or not

So rule seems uninteresting

But $P(\text{Honey}) = 120/1000 = .12$ (hence tea drinkers are far more likely to have honey)

Measure for Association Rules

- So, what kind of rules do we really want?
 - Confidence($X \rightarrow Y$) should be sufficiently high
 - ◆ To ensure that people who buy X will more likely buy Y than not buy Y
 - Confidence($X \rightarrow Y$) > support(Y)
 - ◆ Otherwise, rule will be misleading because having item X actually reduces the chance of having item Y in the same transaction
 - ◆ Is there any measure that capture this constraint?
 - Answer: Yes. There are many of them.

Measures that take into account statistical dependence

$$\text{Lift} = \frac{P(Y | X)}{P(Y)}$$

$$\text{Interest} = \frac{P(X, Y)}{P(X)P(Y)}$$

lift is used for rules while
interest is used for itemsets

$$PS = P(X, Y) - P(X)P(Y)$$

$$\phi - \text{coefficient} = \frac{P(X, Y) - P(X)P(Y)}{\sqrt{P(X)[1 - P(X)]P(Y)[1 - P(Y)]}}$$

Example: Lift / Interest

$$I(A, B) = \begin{cases} = 1, & \text{if } A \text{ and } B \text{ are independent;} \\ > 1, & \text{if } A \text{ and } B \text{ are positively related;} \\ < 1, & \text{if } A \text{ and } B \text{ are negatively related.} \end{cases}$$

	Coffee	<u>Coffee</u>	
Tea	150	50	200
<u>Tea</u>	650	150	800
	800	200	1000

Association Rule: Tea $\rightarrow$ Coffee

Confidence = $P(\text{Coffee}|\text{Tea}) = 0.75$

but $P(\text{Coffee}) = 0.8$

$\Rightarrow \text{lift} = 0.75/0.8 < 1$

$\Rightarrow \text{Interest} = 0.15 / (0.2 \times 0.8) = 0.9375 (< 1, \text{ therefore is negatively associated})$

Example: Lift / Interest

Custo mers	Tea	Honey	...
C1	0	1	...
C2	1	0	...
C3	1	1	...
C4	1	0	...
...			

	<i>Honey</i>	$\overline{Honey}$	
<i>Tea</i>	100	100	200
$\overline{Tea}$	20	780	800
	120	880	1000

Association Rule: Tea $\rightarrow$ Honey

Confidence $\cong P(\text{Honey}|\text{Tea}) = 100/200 = 0.50$

But $P(\text{Honey}) = 120/1000 = .12$ (hence tea drinkers are far more likely to have honey)

$\Rightarrow \text{lift} = 0.50/0.12 = 4.16 > 1$

$\Rightarrow \text{Interest} = 0.1 / (0.12 \times 0.2) = 4.16 (> 1, \text{ therefore is positively associated})$

There are lots of measures proposed in the literature

Measure (Symbol)	Definition
Correlation (ϕ)	$\frac{N f_{11} - f_{1+} f_{+1}}{\sqrt{f_{1+} f_{+1} f_{0+} f_{+0}}}$
Odds ratio (α)	$(f_{11} f_{00}) / (f_{10} f_{01})$
Kappa (κ)	$\frac{N f_{11} + N f_{00} - f_{1+} f_{+1} - f_{0+} f_{+0}}{N^2 - f_{1+} f_{+1} - f_{0+} f_{+0}}$
Interest (I)	$(N f_{11}) / (f_{1+} f_{+1})$
Cosine (IS)	$(f_{11}) / (\sqrt{f_{1+} f_{+1}})$
Piatetsky-Shapiro (PS)	$\frac{f_{11}}{N} - \frac{f_{1+} f_{+1}}{N^2}$
Collective strength (S)	$\frac{f_{11} + f_{00}}{f_{1+} f_{+1} + f_{0+} f_{+0}} \times \frac{N - f_{1+} f_{+1} - f_{0+} f_{+0}}{N - f_{11} - f_{00}}$
Jaccard (ζ)	$f_{11} / (f_{1+} + f_{+1} - f_{11})$
All-confidence (h)	$\min \left[\frac{f_{11}}{f_{1+}}, \frac{f_{11}}{f_{+1}} \right]$